

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Operating Systems ■ CS-3009 ■ Autumn-2023, End Sem

■ 50 Marks ■ Time: 3hrs ■ 2 Pages ■ Answer ALL Questions

NOTE: DO NOT use pencil for answering. Answer all Questions. Questions can be answered in any serial order. However, ensure that all parts of a question are answered sequentially.

1. (a) Consider a system with 'm' CPUs and 'n' processes where $m < n$, then what will be the upper bound and lower bound on number of processes that may be present in the running state, and waiting state respectively? Justify. [2]
(b) Differentiate between a *program*, an *executable*, and a *process*. [2]
(c) Assume every process has 2 seconds of burst time in a system with a single processor. If processes are arriving at the rate of 20 processes per minute, then what is the percentage of time the CPU is idle in the system if it follows SRTF scheduling. [2]
(d) Consider an implementation of counting semaphore, where the waiting list is maintained as stack instead of queue. What would be its impact? Justify your answer. [2]
(e) What are the differences between shared memory IPC and message passing IPC? [2]
2. (a) A computer system has a 36-bit logical address space with a page size of 8K, and 4 bytes per page table entry. Assume that the Page Table stores the mapped frame numbers only. Answer the following. [1+1+4]
 - i. How many pages are in the logical address space?
 - ii. What is the maximum size of addressable physical memory in this system?
 - iii. Would you use a one-level, two-level, or three-level page table? Why? Show the split of logical address into different parts with their respective number of bits.
(b) Consider a system with 3 types of resources: X, Y, and Z; and 3 processes: P0, P1, and P2. The table given below, that presents the current system state, is self explanatory. There are 3 units

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state: [4]

- i. REQ0: P0 requests 0 units of X, 0 units of Y, and 2 units of Z.
- ii. REQ1: P1 requests 2 units of X, 0 units of Y, and 0 units of Z.

Check if the above requests can be granted or not. Justify your answer.

3. (a) Assume that the main memory is divided into 3 frames which are free initially. Consider the following page reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. The system uses LRU (stack based implementation) page replacement algorithm. Show the progressive content of memory frames and the stack for each page reference. [5]

(b) Consider the following program fragment:

[5]

```
void fun(){
    if(a > 0)    P(s1);
    else    P(s2);
    b++;
    P(s3);
    if(b < 0 && a <= 0)    P(s1);
    else if(b >= 0 && a > 0)    P(s2);
    else    P(s4);
    a++;
    V(s4);    V(s3);    V(s2);    V(s1);
}
```

Consider two threads running this fragment of code simultaneously, can there be a deadlock in the following cases? Why, or why not?

► if all the variable are local variables inside fun().

► if all the variables are global variables.

4. (a) What do you mean by thrashing? Does one process thrashing affect only itself, or does it have an effect on the rest of the processes in the system as well? Explain briefly. [2]
- (b) Consider a single-level paging scheme using TLB. TLB access time is 10ns and the main memory access time takes 50ns. Calculate the effective memory access time (in ns) if the TLB hit rate is 90% and there is no page fault. [3]
- (c) What is the difference between a *page* and a *frame*. In paged-segmentation (or segmentation with paging) memory management system, what is structure of the logical address and discuss, with neat diagram, how the MMU translates the logical address to the physical address in such a system. [5]
5. (a) Explain with neat diagram how the OS supports inter-process memory protection so that a process cannot access the address space of any other concurrently executing processes. [3]
- (b) Explain the issues related to contiguous allocation of blocks to files in secondary memory management. [3]
- (c) What is the difference between internal and external fragmentation? Given six memory partitions (fixed size) of 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 357 KB, 210 KB, 468 KB, and 491 KB (in order)? Which algorithm makes the most efficient use of memory? Clearly show the External and Internal fragmentation in case if occurs. [4]